



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## Lab experiment-20

### Experiment 20. Towers of Hanoi

#### Aim

To implement the **Towers of Hanoi problem using Prolog.**

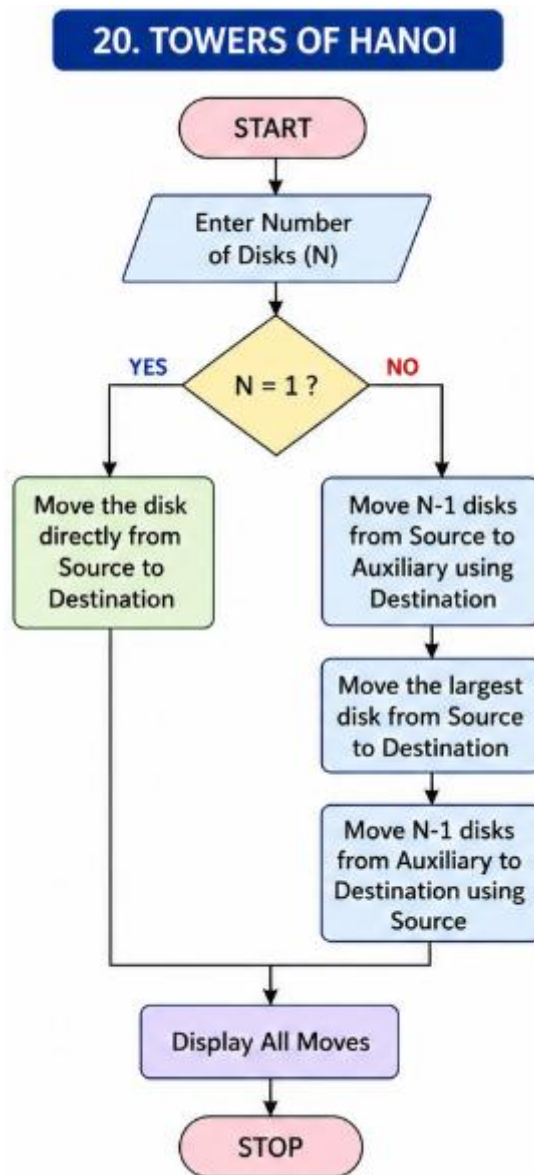
#### Objectives

- To understand recursion in Prolog.
- To solve the Towers of Hanoi problem.
- To generate the required disk movements.

#### Algorithm

1. If there is one disk, move it directly.
2. Move N-1 disks from source to auxiliary.
3. Move the largest disk to destination.
4. Move N-1 disks from auxiliary to destination.

## Flowchart



## Code

hanoi(1, From, To, \_) :-

write('Move disk from '),

write(From),

write(' to '),

write(To),

nl.

hanoi(N, From, To, Aux) :-

$N > 1$ ,

N1 is N - 1,

hanoi(N1, From, Aux,

To),

hanoi(1, From, To, Aux),

hanoi(N1, Aux, To,

From).

main :-

$N = 3$ ,

hanoi(N, left, right,

middle).

:- initialization(main).

## Output

```
HelloWorld.pl ×
hanoi(1, From, To, _) :-
    write('Move disk from '),
    write(From),
    write(' to '),
    write(To),
    nl.
hanoi(N, From, To, Aux) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, From, Aux, To),
    hanoi(1, From, To, Aux),
    hanoi(N1, Aux, To, From).
main :-
    N = 3,
    hanoi(N, left, right, middle).
:- initialization(main).
```

I/O AI Agent

STDIN

Output:

Move disk from left to right  
Move disk from left to middle  
Move disk from right to middle  
Move disk from left to right  
Move disk from middle to left  
Move disk from middle to right  
Move disk from left to right

## Result

Move disk from left to right

Move disk from left to middle

Move disk from right to middle

Move disk from left to right

Move disk from middle to left

Move disk from middle to right

Move disk from left to right

## Conclusion

Thus, the Towers of Hanoi problem was successfully implemented using Prolog.